CHAPTER EIGHT

An overview of fungal glycan-based therapeutics

Dandan Yang[a,*], Zijing Zhou[a,b], Lijuan Zhang[a,*]
[a]Systems Biology and Medicine Center for Complex Diseases, Affiliated Hospital of Qingdao University, Qingdao, China
[b]School of Medicine and Pharmacy, Ocean University of China, Qingdao, China
*Corresponding authors: e-mail address: dandanyang1990@163.com; zhanglj@qduhospital.cn

Contents

1. Introduction 136
2. Structural studies of fugal glycans 136
   2.1 Structure of fungal glycans 136
   2.2 Immunomodulation and anti-tumor effects 137
3. Fungal glycans-based therapeutics 140
   3.1 *G. lucidum* glycan 140
   3.2 Pachyman from *P. cocos* 150
   3.3 Lentinan 150
   3.4 *T. fuciformis* glycan 151
   3.5 Maitake glycan from *G. frondosa* 151
   3.6 *C. versicolor* glycan 152
   3.7 *G. sinense* glycan 152
   3.8 *P. umbellatus* glycan 153
4. Future perspectives 153
Acknowledgments 153
Conflict of interest 153
References 154

Abstract

Edible medicinal mushrooms have been traditionally used for health promotion and longevity in China and other East Asian countries for centuries. Structural and pharmacological studies revealed that fungal glycans show multiple physiological and healthy promoting effects including immunomodulation, anti-tumor, anti-aging, anti-oxidation, hypoglycemic, hypolipidemic, anti-radiation, and other effects. Fungal glycans isolated from different kinds of medicinal mushrooms are partially purified and clinically tested. Without serious safety concerns of mostly glycans from edible mushrooms and/or the cultured mycelium, eight of them are approved by Chinese Food and Drug Administration (SFDA) and used clinically in China since 1980s. In this chapter, 185 independent studies involving in biochemical, pharmacological and clinical studies of fungal glycans

Progress in Molecular Biology and Translational Science, Volume 163
ISSN 1877-1173
https://doi.org/10.1016/bs.pmbts.2019.02.001
© 2019 Elsevier Inc.
All rights reserved.

during the past four decades (1977–2019) from PubMed, CNKI (China National Knowledge Infrastructure) and Wanfang databases are summarized. In future, understanding the fungal glycan-based drugs at molecular biological level would be needed to comprehend the clinical efficacy of glycan-based drugs.

1. Introduction

Glycans, also called polysaccharides, are carbohydrate-based polymers made by all living organisms. Glycans are essential biomolecules serving structure, energy storage and system regulatory purposes. There are plant, animal, and microbial glycans according to their biological sources.

Fungal glycans are microbial glycans isolated from the fruiting body, spore, mycelium, or ferment liquor of fungi. They are natural high molecular weight polymers composed of monosaccharides linked by glycosidic bonds. The unique and complex structures make glycans most information dense biopolymers on earth. More than 200 bioactive glycans have been separated from nearly 300 fungi (more than 70 genera belonging to 51 families) world widely, among which more than 30 fungal glycans with important pharmaceutical value have been identified in China, such as *Ganoderma lucidum* glycan, lentinan from *Lentinus edodes*, *Polyporus* glycan, *Flammulina velutipes* glycan, *Cordyceps sinensis* glycan and *Hericium erinaceus* glycan.[1] In addition to being energy and structural substances, fungi glycans also participate in the regulation of various physiological processes, including enhancing immune function, anti-tumor, hypoglycemic, hypolipidemic, anti-oxidation, anti-radiation, anti-aging, anti-viral and other effects.[2–4] Moreover, glycan-lipid, glycan-peptide and glycan-protein complexes isolated from fungi also have potent biological activities.[2] This chapter will provide the general information of structures and biological activities of the fungal glycans. The molecular mechanisms and clinical applications of the fungi glycan-based drugs will also be discussed.

2. Structural studies of fugal glycans

2.1 Structure of fungal glycans

The structures of fungal glycans are very complex as results of different monosaccharide compositions, glycosidic linkages, molecular weights, and biological activities. Different purification schemes lead to obtaining different glycan fractions. At present, the extraction methods of fungal glycans

mainly include hot water extraction, acid extraction, alkali extraction, enzymatic extraction, ultrasonic assisted extraction and microwave assisted extraction.[3] Taking *G. lucidum* glycans as an example (Table 1), there are 18 different types of *G. lucidum* glycans with different monosaccharide compositions and different glycan linkages being extracted, purified and identified with the molecular weight ranging from 5.2×10^3 to 2.5×10^6 Da,[5–19] indicating that the fungal glycans are not a pure compound.

The two most well-known fungal glycans are β-glucans and mannans. β-Glucans consist of β-glycosidic bond-linked glucose, such as lentinan, *P. cocos* glycan and *C. versicolor* glycan. β-Glucans are the major component of fungal glycans and are the simplest and the most studied fungal glycans. The biologically active fungal β-glucans are those comprising β $(1 \rightarrow 3)$ linked-glucose with side-chains of glucose with β $(1 \rightarrow 6)$ linkage.[2]

Mannans are composed of α glycosidic bond-linked mannoses, especially enriched in α $(1 \rightarrow 3)$ and α $(1 \rightarrow 2)$ linked mannoses, such as *C. sinensis* glycan, *T. fuciformis* glycan and *G. lucidum* glycan.

2.2 Immunomodulation and anti-tumor effects

The spatial structure of glycans, the presence or absence of side chains, the length and connection position of side chains, the branching density and the binding form of glycoside bonds between the chains are all related to the pharmacological activities of fungal glycans.[1] Fungal glycans have inhibition effects on multiple tumors in animal models by enhancing immune function as biological response modifiers (BRM). According to the published studies, β-glucans show significant anti-tumor activities. *Schizophyllum* glycan and lentinan, whose backbones are β $(1 \rightarrow 3)$-D-glucans with a branched degree of 0.33, show strong anti-tumor effects, while β-glycans with other backbone and branching degrees show weaker anti-tumor effects.[20–22]

β-Glucans display immunomodulating and anti-tumor activities by activating the immune cells. β-Glucans can bind to six identified receptors on the cell membrane of immune cells, including complement receptor 3 (CR3), Dectin-1, Fc receptor (FcR), lactosylceramide (LacCer), toll-like receptors 2/6 (TLR2/6) and scavenge receptor (Sc).[23–30] Subsequently, multiple signaling pathways are activated and merged onto important immune regulatory pathways, such as NF-κB pathway. Fungal β-Glucans can activate macrophages, NK cells, APCs and T- and B-lymphocytes, enhance the anti-tumor effect of immune cells, stimulate the secretion of cytokines, activate the complement system, and induce apoptosis of tumor

Table 1 General characteristics of different *Ganoderma lucidum* glycans.

Sources	Extraction methods	Backbone	Major monosaccharide component	Mw	References
Spore	Hot-water extraction	β-Glucan	Glucose	1×10^4	5
		β-Glucan	Glucose	10.3×10^4	6
		β-Glucan	Glucose	1.57×10^5	7
Mycelium	Alkali extraction	β-Glucan	Glucose	–	8
Extracellular metabolite	DEAE-Sephacel and Sephadex G200	α-Galactose	Galactose, mannose, glucose, arabinose, rhamnose	2.2×10^4	9
Fruiting body	Alkali extraction	β-Glucan	Glucose	–	10
		Linear α-glucan	Glucose	–	
	Ultrasound/microwave assisted extraction	β-Glucose	Glucose, galactose	2.5×10^6	11
	Hot-water extraction	Arabinoxyloglucan	Glucose, xylose, arabinose	4×10^4	12
		Heteroglycan	Glucose, galactose, mannose, arabinose, xylose, fucose	–	8
		Heteroglycan	Glucose, galactose, rhamnose	8.3×10^3	13
		β-Glucan	Glucose	6.3×10^4	
		β-Heteroglycan	Glucose, mannose	2.0×10^5	

	α-Galactose	Galactose, glucose, fucose	1.2×10^4	14,15
	α-Glucose			
	α-Galactose	Galactose, glucose, fucose	7×10^3	16
	β-Glucose			
	β-Glucan	Glucose	5.2×10^3	17
	Heteroglycan	Glucose, galactose, mannose	1.54×10^4	
	α-Galactose	Galactose, glucose, fucose	1.12×10^4	18
	β-Glucose			
	Heteroglycan	Glucose, galactose, rhamnose	7.8×10^4	19

cells through activating intracellular signal transduction.[31–34] Even though the anti-tumor effects of mannans are weaker than β-glucans, they still have the anti-tumor efficacies in animal models.[35–38] All published experimental results indicate that fungal glycans can active both the innate and the acquired immune systems to produce anti-tumor immunity, which provides with an important approach for cancer immunotherapy.

Moreover, biological active β-glucans can also be isolated from bacteria, plants and algae. Most of these β-glucans have anti-inflammatory properties, but their anti-inflammatory properties could not be explained by the signal transduction mechanism mentioned above for fungal glycans.

Meanwhile, hypoglycemic and anti-oxidative effects are also common to all fungal glycans. There are many different biological activities and clinical efficacies in different kinds of fungal glycans, which suggests the unique glycan compositions, but not the β-glucans alone, might contribute to the observed effects. Being not pure might be a reasonable explanation of unique biological functions of fungal glycans for fungal glycans by nature are never pure even for the simplest β-glucans.

3. Fungal glycans-based therapeutics

There are eight fungal glycan-based drugs approved by Chinese Food and Drug Administration (SFDA) for clinical use on account of their potent efficacies including *G. lucidum* glycan, *Poria cocos* glycan, lentinan, *Tremella fuciformis* glycan, *Grifola frondosa* glycan, *Coriolus versicolor* glycan, *G. sinense* glycan and *P. umbellatus* glycan. We will briefly review the sources, drug types, fungal glycan contents and clinical applications of the eight fungal glycan-based drugs (Table 2). The biological studies of these fungal glycans are summarized in Table 3.

3.1 *G. lucidum* glycan

The main component of *G. lucidum* glycan is β-glucan, and meanwhile, galactose, xylose, fucose, glucose, mannose and arabinose are present in *G. lucidum* glycan.[7,19,153,154] A *G. lucidum* glycan-based product called "Ji 731 Injection" has been clinically used for treating myopathy in China since 1973. "Polysaccharidum of *G. lucidum* Karst Injection" was approved by SFDA for clinical use in 2000. The major component (no less than 90%) in the drug is *G. lucidum* glycan. This product is mainly used for treating atrophic myotonia, dermatomyositis, polymyositis, facial paralysis and muscular dystrophy, and other diseases such as dyslipidemia.[39–42]

Table 2 General characteristics of eight fungal glycan based-drugs approved by SFDA.

Fungal species	Images	Sources	Drug types	Fungal glycan contents	Clinical applications	References
Ganoderma lucidum		Spore	Injection	*G. lucidum* glycans no less than 90%	Facial paralysis	39–42
					Dyslipidemia	
					Improving endurance of cyclists	
					Dermatomyositis	
					Atrophic myotonia	
					Neurodermatitis	
					Polymyositis	
Poria cocos		Mycelium	Oral solution	Pachyman no less than 84%	Alopecia	43–46
					Chronic pulmonary edema	
					Insomnia	
					Schizophrenia	

Continued

Table 2 General characteristics of eight fungal glycan based-drugs approved by SFDA.—cont'd

Fungal species	Images	Sources	Drug types	Fungal glycan contents	Clinical applications	References
Lentinus edodes		Fruiting body	Injection	Lentinan no less than 85%	Lung cancer	47–73
					Gastrointestinal cancers	
					Liver cancer	
			Capsule	Lentinan	Ovarian cancer	
					Cervical cancer	
			Tablet	Lentinan no less than 40%	HIV-positive	
					Hepatitis	
					Malignant pleural effusion	
Tremella fuciformis		Fruiting body	Capsule	*Tremella* glycan no less than 60%	Chronic hepatitis	74–83
					Mycoplasma pneumonia	
					Leukopenia	
					Relieving subjective cognitive impairment	

Grifola frondosa		Mycelium	Capsule	Maitake glycan no less than 40%	Breast cancer	84–88
					Primary liver cancer	
					Gastrointestinal cancers	
					Polycystic ovary syndrome	
					Impaired glucose tolerance	
					Cosmetics	
Coriolus versicolor		Mycelium	Capsule	Krestin no less than 35%	Liver cancer	89–94
					Lung cancer	
					Gastric cancer	
			Tablet	Krestin	Colorectal cancer	
					Acute nonlymphocytic leukemia	
			Oral solution		Hepatitis	
					Hyperlipidemia	
			Granule		Chronic bronchitis	
					Ulcer	

Continued

Table 2 General characteristics of eight fungal glycan based-drugs approved by SFDA.—cont'd

Fungal species	Images	Sources	Drug types	Fungal glycan contents	Clinical applications	References
Ganoderma sinensis		Fruiting body	Tablet	*G. sinensis* glycan no less than 83%	Leukopenia	95,96
					Mushroom poisoning	
Polyporus umbellatus		Fruiting body	Capsule	*Polyporus* glycan no less than 40%	Hepatitis B	97–105
					Bladder cancer	
					Gastric cancer	
			Injection	*Polyporus* glycan no less than 90%	Liver cancer	
					Leukopenia	
					Cervical cancer	

Table 3 The biological activities of the eight fungal glycans.

Glycans	Effects	Models	Biological activities	References
Ganoderma lucidum glycans	Immunomodulation	BALB/c mice	Promoting dinitrochlorobenzene-induced delayed-type ear swelling in mice significantly	7
		BALB/c mice	Increasing the number of peripheral blood WBC and PLT and IFN-γ mRNA in spleen lymphocytes remarkably in immunosuppressed mice	106
	Anti-oxidation	_	Exhibiting the scavenging effects on hydroxyl radicals, ABTS radical and DPPH free radical in a concentration-dependent manner	107
		SD rats	Increasing serum MDA level and decreasing SOD activity significantly	108
		Wistar rats	Improving the levels of serum and gastric tissue SOD, CAT and GSH-Px toward the control values in a dose-dependent manner	109
	Anti-tumor	HL-60 cells	Blocking the extracellular signal-regulated kinase/MAPK signaling pathway, simultaneously activated p38 and JNK MAPK pathways, and therefore regulated their downstream genes and proteins, including p53, c-myc, c-fos, c-jun, Bcl-2, Bax, cleaved caspase-3 and cyclin D1	110
		BALB/c mice	Promoting the serum level of IFN-γ, TNF-α, and nitric oxide, enhancing nature killer cell (NK cell) cytotoxic activity	111
		C57BL/6J mice	Increasing significantly the expression of both TNF-a and IFN-γ (at both mRNA and protein levels) in splenocytes in a dose-dependent manner, and increased cytotoxic T lymphocyte cytotoxicity and NK activity	112
		Nude mice	Enhancing the inhibitory effect of cisplatin on tumor growth and angiogenesis in T24 tumor-bearing nude mice	113

Continued

Table 3 The biological activities of the eight fungal glycans.—cont'd

Glycans	Effects	Models	Biological activities	References
	Hyperglycemic	Wistar rats	Exerting noticeable hypoglycemic and hypolipidemic effects ($P<0.05$) in diabetic rats	114
		Wistar rats	Reducing blood fat and blood glucose in diabetes rats when combined with sports	115
Pachymaran	Anti-tumor and immunomodulation	U937 and HL-60 cells	Elevating the levels of cytokines of IFN-γ and TNF-α to inhibit the growth and induce the differentiation of the tumor cells	116
		C57BL/6 mice	Reducing the number of lung surface metastases and increasing the expression of CD11b mRNA in peripheral white blood cells	117
		BALB/c mice and HL-60 cells	Exhibiting antitumor activities *in vivo* and *in vitro*	118
		Rats	Inhibiting tumor cells growth and increased Serum SOD, CAT and GPx activities significantly	119
	Anti-oxidation	_	Carboxymethyl-sulfated pachymaran showed the highest antioxidant abilities on ferric-reducing power, hydroxyl radicals scavenging, superoxide radicals scavenging and lipid peroxidation inhibition	120
		Kunming mice	Improving the superoxidase dismutes capacity and the total anti-oxidant capacity in the spleen or thymus of PCV2-induced mice	121

Lentinan	Immunomodulation	Mice	Increasing production of cytokine in immune cells	122,123
		Chicken	Enhancing serum antibody titer and promote lymphocyte proliferation	124
		Wistar rats	Improving bactericidal ability of peritoneal and alveolar macrophages	125
		Guinea pigs	Inducing high level of alveolar macrophage activation	126
		Mice	Inducing TNF-α secretion of murine macrophages	127
		Rats	Inducing long-term potentiation in the rat dentate gyrus	128
Tremella glycan	Immunomodulation	BALB/c mice	Reversing the influences of Tregs on $CD4^+$ T cells proliferation and polarization and declining the level of IL-10 and mortality in burn sepsis mice	129
		BALB/c mice	Decreasing the secretion of IL-10 from $CD4^+$ T cells	130
		BALB/c mice	Inhibiting the abnormal activities of $CD4^+CD25^{high}$Tregs in burn sepsis mice *via* inhibiting IL-10 secretion	131
	Anti-oxidation	_	Exhibiting the scavenging effects on hydroxyl radicals and DPPH free radical in a concentration-dependent manner	132
		_	Degrading samples with lower molecular weight possessed the higher antioxidant activities	133

Continued

Table 3 The biological activities of the eight fungal glycans.—cont'd

Glycans	Effects	Models	Biological activities	References
	Anti-aging	HDF-n cells	Inhibiting of p16, p21, p53 and caspase-3 expression, and activation of kinase 1 extracellular-signal regulated kinase and Akt serine/threonine and protecting fibroblasts *via* SIRT1	134
	Hypoglycemic	Wistar rats	Reducing cholesterol, triglyceride, GPT, urea and increasing HDL cholesterol	135
Maitake glycan	Immunomodulation and anti-tumor	HepG2 cells	Inhibiting notch1 expression, IκB-α degradation and NF-κB/p65 translocation from cytoplasm into nucleus and activating of caspase-3 and caspase-8	136
		SMMC-772 cells	The upregulation of BAX, downregulation of Bcl-2, activation of poly-(ADP-ribose)-polymerase (PARP), and the release of cytochrome *c* were observed in cells	137
		RAW264.7 cells	Increased the expression of CD4+ and CD8+ splenic T lymphocytes and IL-1β, IL-2, IL-6 and IFN-γ in splenocyte	138
		MCF-7 cells	Stimulating the immune system and activating certain cells that attack cancer, including macrophages, T-cells, and natural killer cells	139
	Anti-oxidation	MRC-5 cells	Better antioxidant shown by the lower EC50 values of DPPH scavenging ability, ferric-reducing antioxidant power and ferrous ion-chelating ability	140

	Hypoglycemic	HepG2 cells	Activating the insulin receptor protein (IRS) in the cell membrane and increasing phosphorylated-AktSer473 expression	141
		SD rats	Significantly decreasing the levels of FSG, FSI and HOMA-IR, the levels of FSG, FSI and HOMA-IR compared with a diabetic control group	142
	Hepatoprotective	L-02 cells	Decreasing the over-active of CYP2E1, TNF-α and ROS fluorescence induced by CCL4	143
	Anti-hypertension	SD rats	Significantly lowering systolic blood pressure (SBP) in diabetic SD rats	144
Krestin	Anti-tumor	Rat	Suppress metastasis induced by hepatic I/R	145
		Rat	Inhibit bone metastasis of breast cancer	146
Ganoderma sinensis glycan	Immunomodulation	Kunming mice	Enhancing levels of IL-2, IL-3, IL-4, interferon γ, TNFα and IL-2R	147
		Mice	Stimulating the release of NO by macrophages and the proliferation of mouse lymphocytes	148
Polyporus glycan	Immunomodulation and anti-tumor	C3H/HeN mice	Enhancing TNF-α, IL-1β, and NO production	149
		Fisher-344 rats	Inhibiting bladder carcinogenesis, which may be associated with upregulation of GSTPi and NQO1 in the bladder	150
	Kidney protection	SD rats	Preventing the progression of renal injury and the subsequent renal fibrosis in aristolochic acid nephropathy	151
		SD rats	Down-regulating AQP2 by down-regulating V(2)R	152

Mouse- and rat-based animal studies indicate that *G. lucidum* glycans can regulate the immune function.[7,106] *G. lucidum* glycans also have potent anti-oxidative,[107–109] anti-tumor[110–113] and hyperglycemic activities.[114,115]

3.2 Pachyman from *P. cocos*

Pachyman is isolated from the mushroom of *P. cocos*. The major components are β-glucan with β-(1 → 3) linked glucose backbone and β-(1 → 6)-linked glucose side chains with different combinations of arabinose, xylose, glucose, fucose, galactose and mannose.[155–157]

Pachyman-based product "Compound Polysaccharide Oral Solution" was developed in the 1970s. "Polysaccharidum of *P. cocos* Oral Solution" was approved by SFDA in 2006. The major component (no less than 84%) in the patented product is pachyman. This drug is clinically used for treating alopecia, chronic pulmonary edema, insomnia and schizophrenia.[43–46] Moreover, it prevents tumor metastasis through regulation of the immunity in cancer cell-based animal models.[116,117] Pachyman exhibits antitumor activities both *in vitro* and *in vivo*.[118,119,158] Both mouse- and rat-based animal studies demonstrate the potent anti-oxidative effects of pachyman.[120,121,159,160]

3.3 Lentinan

Lentinan is isolated from the mushroom *L. edodes*. The primary structure of β-glucan in lentinan is consisted of β-(1–3)-glucose backbone with two (1–6)-β-glucose branches of each five glucose units.[161,162] Lentinan-based drugs are available as injections, capsules and tablets in China. Lentinan is clinically used for cancer treatment both in China and Japan. According to the clinical studies published in and outside of China, lentinan-based drugs are used for the treatment of various cancers, including lung cancer,[47–58] gastric cancer,[59–63] colorectal cancer[64–68] and other cancers.[69,70] In addition, lentinan-based drugs are also used for treating HIV, hepatitis and malignant pleural effusion.[71–73]

Lentinan is approved as an adjuvant for stomach cancer therapy in Japan in 1985. As a biological response modifier, lentinan inhibits the tumor growth by regulating the immune system through activating immune cells, promoting the T- and B-lymphocyte proliferation and enhancing the activities of NK cells in mouse-, rat-, chicken- and guinea pig-based animal models.[122–128] These studies also demonstrate that lentinan stimulates the productions of different cytokines.

3.4 *T. fuciformis* glycan

The main structure of *T. fuciformis* glycan consists of a linear (1 → 3)-linked α-D-mannose backbone chain with highly branched β-D-xylose, α-D-mannose and β-D-glucuronic acid in the side chains.[163,164] Other glycans may also contain xylose, mannose, glucose, galactose and glucuronic acid.[165,166]

T. fuciformis glycan-based product is "*Tremella* Polysaccharide Enteric-coated Capsules" which was approved by SFDA in 2002. The major component (more than 60%) in the product is *T. fuciformis* glycan.[167] This drug is authorized for clinical use in treating patients with leukopenia caused by chemotherapy, radiotherapy and other treatments. It is also authorized as an adjuvant in treating chronic hepatitis and chronic active hepatitis. According to the published studies, *T. fuciformis* glycan-based drugs have good efficacy for patients with chronic hepatitis,[74–76] mycoplasma pneumonia[77] and leukopenia induced by chemotherapy, radiotherapy and other causes.[78–82] In recent years, a study indicates that this drug can also help to relieve the subjective cognitive impairment.[83]

Mouse-based animal studies revealed that *T. fuciformis* glycan can promote the immune function significantly in mouse model of sepsis by regulating the cellular immunity and cytokine secretions.[129–131] In addition, *T. fuciformis* glycan shows great anti-oxidant, anti-aging and hypoglycemic effects in mouse- and rat-based animal models.[132–135,168–171] Meanwhile, studies showed that *T. fuciformis* glycan has the neuroprotective and neurotrophic effects in animal models.[172–174]

3.5 Maitake glycan from *G. frondosa*

Maitake glycan is isolated from the edible and medicinal fungus *G. frondosa*. β-Glucans is the main component of maitake glycan and the bioactive β-glucan fraction termed D-fraction is very promising in enhancing the immune system whether by oral administration or by injection. Other fungal glycans with anti-tumor effects are mainly composed of β-(1 → 3)-glucan as a main chain with β-1,6 branches, In contrast, D-fraction is composed of β-(1 → 6)-glucan as a main chain with β-1,3 branches.[175]

Maitake glycan-based drugs approved by SFDA are available as capsules named "Hui Shu Hua Capsules," with no less than 40% maitake glycans isolated from cultured mycelium of *G. frondosa*. This drug is clinically used for treating patients with cancer,[84,85] polycystic ovary syndrome[86] and impaired glucose tolerance conditions[87] according to the published studies. In addition, maitake glycans are also used for cosmetic purpose.[88]

Mouse- and rat-based animal studies demonstrate that maitake glycans can inhibit the tumor growth by regulating cytokine productions and by activating immune cells. Maitake glycan induces cell apoptosis through a notch1/NF-B/p65-mediated caspase pathway.[136–139] Moreover, maitake glycans also show potent anti-oxidative, hypoglycemic, hypo-systolic blood pressure and hepatoprotective effects.[140–144]

3.6 *C. versicolor* glycan

The *C. versicolor* glycan is mainly composed of β-glucan with a backbone of β-(1–3)-glucose with (1–6)-β-glucose branches.[176] Krestin is a protein bound glycan isolated from cultured mycelium of *C. versicolor*. The molecular weight and structures of the *C. versicolor* glycan and krestin are different in different strains and are variable in different culture conditions.[177]

So far, 12 kinds of *C. versicolor* glycan-based drugs are authorized by SFDA for clinical use in China including tablets, capsules, oral solutions and granules.[178] The published clinical studies showed that *C. versicolor* glycan-based drugs can be used for the treatment of various cancer like liver cancer, lung cancer, gastric cancer, colorectal cancer and acute nonlymphocytic leukemia.[89–94] In addition, these drugs have good efficacies in treating hepatitis, hyperlipidemia, chronic bronchitis and ulcer.[179–181]

Three mechanisms are proposed to explain the anti-tumor effects of *C. versicolor* glycan.[145,146,182] Firstly, the glycan can activate T-cells, B-cells, NK-cells, cytotoxic T-cells, antigen-presenting cells and tissue macrophages accompanied by releasing a variety of cytokines and growth factors. Secondly, it can improve innate immune-competence by inhibiting the production and neutralizing the immunosuppressive substances that are increased in cancer. Thirdly, the glycan has direct antitumor effects by inhibiting cancer cell growth.

3.7 *G. sinense* glycan

The major component of *G. sinensis* glycans are β-glucans, water-soluble heteroglycans and glycan-protein complex, and glucose, xylose, arabinose, fucose, mannose and galactose are all detected in the glycans. The molecular weight of the glycans ranges from 10^3 to 10^6 Da.[183]

G. sinensis glycan-based capsules are authorized by SFDA for clinical use in China. Reported clinical studies demonstrated that this drug stimulates leukocyte productions in leukopenia patients and neutralizes mushroom

poisoning.[95,96] Further studies showed that *G. sinensis* glycan can significantly enhance the immune function by promoting the proliferation of T cells and the secretion of cytokines.[147,148]

3.8 *P. umbellatus* glycan

The major bioactive of *P. umbellatus* glycan is a β-glucan with a (1–3)-β-glucose backbone and (1–6)-β-glucose side chains with a molecular weight of approximately 1.6×10^5 Da, which is isolated from the cultured mycelium.[21,184]

Polyporus glycan-based drugs are available as capsules and injections in China. The published clinical studies indicate that capsules are used for treating hepatitis B. The injections are effective in reducing the recurrence of bladder cancer.[97–102] *Polyporus* glycans are also used alone or together with chemotherapy drugs to treat patients with gastric cancer, liver cancer, leukemia and cervical cancer.[103–105] Mouse-based animal studies indicate that *P. umbellatus* glycans also have potent anti-tumor and kidney protective effects.[149–152]

4. Future perspectives

Compared with the conventional small molecule- or protein-based drugs, the advantages of glycan-based drugs are their broad spectrum of therapeutic properties, relatively low toxicity, relatively low costs and less drug-resistant issues. But there are some problems urgently needed to be solved before fungal glycan-based drugs are accepted by governments and clinicians worldwide, such as how to standardize quality, composition, purity of the highly dispersed glycan molecules, how to comprehend the pharmacodynamics of fungal glycan-based drugs at molecular level and how to perform reliable pharmacokinetic studies. Thus, developing novel structural characterization tools and reliable biological assays might be critical in understanding the fungal glycan-based drugs in future.

Acknowledgments

This research was supported by the Natural Science Foundation of China (Grant 81672585), Key Technology Fund of Shandong Province (Grant 2016ZDJS07A07), the Taishan Scholar Fellowship, and the "Double First Class Fund" of Shandong Province in China to L.Z and the Postdoctoral fellowship of Qingdao Research Project, China to D.Y.

Conflict of interest

The authors declare no conflict of interest.

References

1. Chen C, Jin W, Li R, Bao S. Advances in pharmacological action of fungal polysaccharides. *Li Shizhen Med Materia Med Res*. 2006;17(6):933–934.
2. Zhou Z, Han Z, Zeng P, et al. Chinese FDA approved fungal glycan-based drugs: an overview of structures, mechanisms and clinical related studies. *Transl Med*. 2014;4(4):141.
3. Guo T, Yan X, Hu X, Liang N, Chen P. Research advancement of fungi polysaccharides. *Prog Mod Biomed*. 2013;13(18):3578–3583.
4. Chen G, Ren Y. Research status and development prospect of edible and medicinal fungal polysaccharides were analyzed. *For All Health*. 2015;9(4):288.
5. Bao X, Liu C, Fang J, Li X. Structural and immunological studies of a major polysaccharide from spores of *Ganoderma lucidum* (Fr.) karst. *Carbohydr Res*. 2001;332:67–74.
6. Dong Q, Wang Y, Shi L, et al. A novel water-soluble β-D-glucan isolated from the spores of *Ganoderma lucidum*. *Carbohydr Res*. 2012;353:100–105.
7. Wang Y, Liu Y, Yu H, et al. Structural characterization and immuno-enhancing activity of a highly branched water-soluble β-glucan from the spores of *Ganoderma lucidum*. *Carbohydr Polym*. 2017;167:337–344.
8. Sone Y, Okuda R, Wada N, Kishida E, Misaki A. Structures and antitumor activities of the polysaccharides isolated from fruiting body and the growing culture of mycelium of *Ganoderma lucidum*. *Agric Biol Chem*. 1985;49:2641–2653.
9. Li Y, Fang L, Zhang K. Structure and bioactivities of a galactose rich extracellular polysaccharide from submergedly cultured *Ganoderma lucidum*. *Carbohydr Polym*. 2007;68:323–328.
10. Chen J, Zhou J, Zhang L, Nakamura Y, Norisuye T. Chemical structure of the water-insoluble polysaccharide isolated from the fruiting body of *Ganoderma lucidum*. *Polym J*. 1998;30(10):838–842.
11. Huang S, Li J, Li Y, Wang Z. Purification and structural characterization of a new water-soluble neutral polysaccharide GLP-F1-1 from *Ganoderma lucidum*. *Int J Biol Macromol*. 2011;48:165–169.
12. Miyazak T, Nishijima M. Structural examination of a water-soluble, antitumor polysaccharide of *Ganoderma lucidum*. *Chem Pharm Bull*. 1981;29:3611–3616.
13. Bao X, Wang X, Dong Q, Fang J, Li X. Structural features of immunologically active polysaccharides from *Ganoderma lucidum*. *Phytochemistry*. 2002;59:175–181.
14. Ye L, Zhang J, Zhou K, et al. Purification, NMR study and immunostimulating property of a fucogalactan from the fruiting bodies of *Ganoderma lucidum*. *Planta Med*. 2008;74:1730–1734.
15. Ye L, Zhang J, Ye X, et al. Structural elucidation of the polysaccharide moiety of a glycopeptide (GLPCW-II) from *Ganoderma lucidum* fruiting bodies. *Carbohydr Res*. 2008;343(4):746–752.
16. Ye B, Zhang J, Yang Y, et al. Structural characterisation of a heteropolysaccharide by NMR spectra. *Food Chem*. 2009;112(4):962–966.
17. Liu W, Wang H, Pang X, Yao W, Gao X. Characterization and antioxidant activity of two low-molecular-weight polysaccharides purified from the fruiting bodies of *Ganoderma lucidum*. *Int J Biol Macromol*. 2010;46(4):451–457.
18. Ye L, Li J, Zhang J, Pan Y. NMR characterization for polysaccharide moiety of a glycopeptide. *Fitoterapia*. 2010;81(2):93–96.
19. Pan D, Wang L, Chen C, et al. Structure characterization of a novel neutral polysaccharide isolated from *Ganoderma lucidum* fruiting bodies. *Food Chem*. 2012;135(3):1097–1103.
20. Synytsya A, Novak M. Structural diversity of fungal glucans. *Carbohydr Polym*. 2013;92(1):792–809.

21. Zong A, Cao H, Wang F. Anticancer polysaccharides from natural resources: a review of recent research. *Carbohydr Polym*. 2012;90(4):1395–1410.
22. Zhang M, Cui S, Cheung P, Wang Q. Antitumor polysaccharides from mushrooms a review on their isolation process, structural characteristics and antitumor activity. *Trends Food Sci Technol*. 2007;18(1):4–19.
23. Kariya Y, Okamoto N, Fujimoto T, et al. Lysis of fresh human tumor cells by autologous peripheral blood lymphocytes and tumor-infiltrating lymphocytes activated by PSK. *Jpn J Cancer Res*. 1991;82(9):1044–1050.
24. Nio Y, Shiraishi T, Tsubono M, et al. In vitro immunomodulating effect of protein-bound polysaccharide, PSK on peripheral blood, regional nodes, and spleen lymphocytes in patients with gastric cancer. *Cancer Immunol Immunother*. 1991;32(6): 335–341.
25. Vanky F, Wang P, Klein E. The polysaccharide K (PSK) potentiates in vitro activation of the cytotoxic function in human blood lymphocytes by autologous tumour cells. *Cancer Immunol Immunother*. 1992;35(3):193–198.
26. Ebina T, Kohya H. Antitumor effector mechanism at a distant site in the double grafted tumor system of PSK, a protein-bound polysaccharide preparation. *Jpn J Cancer Res*. 1988;79(8):957–964.
27. Tsurus S, Nomoto K. Effects of PSK on specific tumor immunity to syngeneic tumor cells. *J Clin Lab Immunol*. 1983;10(4):215–219.
28. Alarra I, Collado A, Garrido F. Protein bound polysaccharide PSK abrogates more efficiently experimental metastases derived from H-2 negative than from H-2 positive fibrosarcoma tumor clones. *J Exp Clin Cancer Res*. 1997;16(4):373–380.
29. Baba N. The enhancement of tumoricidal activities of macrophages by protein-bound polysaccharide in tumor bearing mice. *Biotherapy*. 1990;4:123–128.
30. Kato H. Tumor inhibitory effect of polymorphonuclear leukocytes (PMN) induced by PSK in the peritoneal cavity of tumor-bearing mice. *J Kyoto Pref Univ Med*. 1987;96:927–937.
31. Wang S, Wang K, Chen D. Progress on immunological regulation effects and mechanisms of edible fungus polysaccharides. *Prog Vet Med*. 2012;33(11):104–108.
32. Xing X, Cui J, Guo H, Qian A. Advances in immunomodulation of fungal polysaccharides. *Heilongjiang Anim Sci Vet Med*. 2012;10:26–28.
33. Li L, Zhao G. Bioactivity of medicinal fungi polysaccharides for anti-tumor immunity. *Chin J Cancer Prev Treat*. 2015;22(14):1156–1160.
34. Wang L, Chen X, Wang H, Yang G, Kong X. Progress on anti-tumor activity of compound of medicinal fungus polysaccharides. *Lett Biotechnol*. 2016;27(6):888–891.
35. Wang C, Cui L, Zhang Z, et al. Study on the regulation of tumor immune response by oral Chinese polysaccharides. *China J Chin Materia Med*. 2016;41(10):1965–1971.
36. Yan Y, Xu W, Shen X, Wang D, Zhao Y, Zhou Z. Influence of *Tremella fuciformis* Berk spores polysaccharide A—BTF to chromosome aberration and micronucleus rate of marrow cells in mice irradiated by 137Cs γ-ray. *Chin J Radiol Health*. 2014;23(1):54–56.
37. Wang X, Yang P, Kong L, Wang Q. Protection of phosphate esterification polysaccharides from *Tremella fuciformis* on hematopoietic function in radiation-injured mice. *Chin Tradit Herb Drug*. 2013;44(13):1811–1813.
38. Han Y, Shen X, Xu W, et al. Radioprotective effect of polysaccharides isolated from *Tremella fuciformis* in mice. *Chin J Radiol Health*. 2012;21(2):132–133.
39. Liu M, Ju C, Zhu Y, Zhu Y. The assisting effect and mechanism of *Ganoderma lucidum* polysaccharides in treatment of depression. *Prog Mod Biomed*. 2014;14(4):656–662.
40. Zhu L. Efficacy GLPSS injection combined with glucocorticoid treatment facial paralysis. *Cap Med*. 2010;22:27.

41. Rossi P, Buonocore D, Altobelli E, Brandalise F, Savino V, Marzatico F. Improving training condition assessment in endurance cyclists: effects of *Ganoderma lucidum* and ophiocordyceps sinensis dietary supplementation. *Evid Based Complement Alternat Med*. 2014;2014:979613.
42. Hu M, Zeng W, Tomlinson B. Evaluation of a crataegus-based multiherb formula for dyslipidemia: a randomized, double-blind, placebo-controlled clinical trial. *Evid Based Complement Alternat Med*. 2014;2014:365742.
43. Zhang L, Wang S. Clinical effect of Poria in treating heart failure edema in patients with chronic pulmonary heart disease (54 cases). *J Pract Tradit Chin Intern Med*. 2004;18(2):164.
44. Fan G. High-dose Poria in treatment of insomnia (24 cases). *Res Tradit Chin Med*. 2006;19(2):35–36.
45. Liu C. Poria in treatment of alopecia (23 cases). *J Hubei Tradit Chin Med*. 1996;1:58.
46. Zhang Y, Luo K, Zhao E, Li Q, Wan S, Liu Z. Changes in IgA and serum ceruloplasmin in chronic schizophrenia with treatment of Poria. *Shanxi J Med*. 1982;11(5):14–15.
47. Yin H, Lin Y. Efficacy of synergistic combination of lentinan and GP regimen in stage IV lung squamous cell carcinoma. *Oncol Prog*. 2016;14(7):687–689.
48. Zhang H. Clinical analysis of lentinan combined with chemotherapy for elderly patients with advanced non-small cell lung cancer. *Shenzhen J Integr Tradit Chin West Med*. 2016;26(4):44–45.
49. He D, Li J, Chen Z. Recent and long-term effect observation of lentinan combined with EP chemotherapy in the treatment of small cell lung cancer. *J Huaihai Med*. 2015;33(6):598–599.
50. Ma J. Efficacy of lentinan combined with NP regimen in treatment of patients with advanced non-small cell lung cancer and its effect on immunologic function. *Chin J Biochem Pharm*. 2014;34(8):154–156.
51. Li C, Zhu T, Lu X. Clinical analysis of the application of lentinous edodes polysaccharides combined with chemotherapy in the treatment of patients with advanced lung cancer. *Mod Diagn Treat*. 2016;27(9):1614–1616.
52. Zhang Y, Yu J, Xiang Y, Liu Y. Clinical efficacy and safety of lentinan combined with chemotherapy for elderly patients with advanced non-small cell lung cancer. *Anti-tumor Pharm*. 2015;5(3):222–225.
53. Song Q, Zhu J. Effect of mushroom polysaccharide injection on serum tumor markers and clinical effect in patients with stage III and IV SCLC. *J China Prescrip Drug*. 2016;14(7):12–13.
54. Dai D. Recent and long-term effect and adverse reaction of lentinan combined with GP chemotherapy in the treatment of non-small cell lung cancer. *Mod J Integr Tradit Chin West Med*. 2014;23(15):1627–1728.
55. Wu X. Clinical analysis of lentinan combined with chemotherapy in treatment of advanced lung cancer. *Chin Arch Tradit Chin Med*. 2015;33(2):456–458.
56. Zhang Y, Xie C, Kang Z, Zeng C, Huang Z. Clinical observation on GP chemotherapy combined with lentinan to treat metaphase or terminal non-small cell lung cancer. *J Qiqihar Uni Med*. 2014;35(2):164–165.
57. Xin W, Shi L, Zhu Y. Efficacy of lentinan combined with NP chemotherapy in treatment of patients with advanced non-small cell lung cancer and its effect on immunologic function. *Mod J Integr Tradit Chin West Med*. 2016;25(30):3349–3351.
58. Wu Y, Xie M. The influence of lentinan injection on curative effect and quality life of patients with medium and advanced lung cancer after chemotherapy. *Guid J Tradit Chin Med Pharm*. 2015;21(22):2–27.
59. Zhao W. Efficacy of lentinan combined with chemotherapy in treatment of 78 cases with advanced gastric cancer. *J Med Forum*. 2014;35(4):118–119.

60. Wang Z, Wu Q. Effect observation of lentinan adjuvant chemotherapy in the treatment of patients after radical resection of gastric cancer. *Shandong Med J*. 2016;56(26):46–48.
61. Lin B, Dai Y. Effects of lentinan in combination with FOLFOX4 regimen on serum vascular endothelial growth factor, interleukin-10 and quality of life in patients with medium and advanced gastric cancer. *World Chin J Digestol*. 2015;23(33):5372–5376.
62. Zhou N, Jiao J. Effect analysis of lentinan combined with XELOX chemotherapy in the treatment of advanced gastric cancer. *Contemp Med Symp*. 2016;14(8):144–146.
63. Li X, Niu M, Zhang M, Zhang R. Immune and clinical effect of lentinan combined with chemotherapy in the treatment of patients with gastric cancer. *Shaanxi J Tradit Chin Med Mod Dist Educ China*. 2015;44(12):1624–1626.
64. Dong X, Li Q. Effect of lentinan injection combined FOLFOX for advanced colon cancer. *Pract J Cancer*. 2015;30(1):136–138.
65. Wang X, Yu X, Shen Z, Luo C. Clinical effect of lentinan combined with chemotherapy for patients with advanced colon cancer. *China Med Her*. 2014;11(17):65–69.
66. Liu Z, Feng Q, Hu C. The curative effect of mushrooms polysaccharide combined with XELOX chemotherapy on advanced colorectal cancer. *Jilin Med J*. 2015;36(15): 3234–3236.
67. Zhang J. Clinical observation on lentinan combined with chemotherapy in the treatment of postoperative patients with colorectal cancer. *Chin Foreign Med Res*. 2015;13(27):32–33.
68. Zhang Y, Ni T, Li Y, Zhang H, Li X, Xue Y. Clinical efficacy and safety of lentinan combined with chemotherapy in elderly patients with colon-rectum cancer. *Med Innov China*. 2015;12(6):16–19.
69. Long A. Curative effect of lentinan combined with cisplatin and paclitaxel on elderly patients with ovarian cancer and ascites. *Clin Med China*. 2016;36(6):5–8.
70. Xiang D, Luo R. Effect of lentinan on the efficacy of docetaxel combined with cisplatin in stage IV cervical cancer. *Oncol Prog*. 2016;14(8):808–810.
71. Wang J. Clinical observation of lentinan in treatment of 60 cases of hepatitis B. *J Chin Med*. 1994;16(5):23–24.
72. Chang X, Shao P, Huang X, Xu W. Clinical research on cisplatin combined lentinan in local therapy of malignant pleural effusion. *New Med*. 2013;44(12):840–842.
73. Gordon M, Bihari B, Goosby E, et al. A placebo-controlled trial of the immune modulator, lentinan, in HIV-positive patients: a phase I/II trial. *J Med*. 1998;29(5–6): 305–330.
74. Guo Z, Tan X. The curative effect of Tremella polysaccharide enteric-coated capsules in preventing and treating acute and chronic liver injury after blood tumor chemotherapy. *Infect Dis Inf*. 2005;18:71.
75. Xiong H, Wang J, Liu L, Zhu S. Clinical observation on 45 cases of chronic hepatitis by the treatment with *Tremella fuciformis* polysaccharides. *Antibiotics*. 1985;10(6):363–365.
76. Li Q, Huang C, Jiao L, Pan S. Clinical study on the treatment of chronic active hepatitis with tremella polysaccharide enteric-coated capsules. *Infect Dis Inf*. 2006;19(4): 201–202.
77. Zhao X. Clinical observation on the treatment of mycoplasma pneumonia by the combination of tremella polysaccharide enteric-coated capsules and azithromycin. *Chin Med Mod Dist Educ China*. 2009;7(6):111.
78. Liu C. Comparison of trichosporose and hematopoiesis in the treatment of mesimazole—induced leukopenia. *Clin Med China*. 2001;17(12):934.
79. Huang Z, Bai J, Wu M. 58 cases of leukopenia treated by tremella polysaccharides. *Chongqing Med*. 1982;5:14–16.
80. Cheng G. Clinical study of preventive and curative effects of *Tremella fuciforums* Berk on tumor patients with leukopnia. *Antibiotics*. 1982;7(1):52–55.

81. Wang Y, Sun M, Li X. Clinical observation on the treatment of interferon-induced leukopenia by tremella polysaccharide enteric-coated capsules. *Hebei Med J*. 2011;33(3):411.
82. Pang L. Clinical study about parenteral solution of tremella polysaccharide improving quality of life of patients with liver cancer during chemotherapy. *J Hubei Uni Chin Med*. 2014;16(4):85–86.
83. Ban S, Lee SL, Jeong HS, et al. Efficacy and safety of *Tremella fuciformis* in individuals with subjective cognitive impairment: a randomized controlled trial. *J Med Food*. 2018;21(4):400–407.
84. Deng G, Lin H, Seidman A, et al. A phase I/II trial of a polysaccharide extract from *Grifola frondosa* (Maitake mushroom) in breast cancer patients: immunological effects. *J Cancer Res Clin Oncol*. 2009;135(9):1215–1221.
85. Liu A, Zang L, Sun Q. Clinical observation of effect on *Grifola frondosa* amylose against tumor. *J Shandong Inst Light Ind: Sci Technol*. 2008;22(2):43–45.
86. Chen J, Tominaga K, Sato Y, Anzai H, Matsuoka R. Maitake mushroom (*Grifola frondosa*) extract induces ovulation in patients with polycystic ovary syndrome: a possible monotherapy and a combination therapy after failure with first-line clomiphene citrate. *J Altern Complement Med*. 2010;16(12):1295–1299.
87. Ren H, Wang H, Yuan F, Zhang J. Clinical observation of maitake capsules in the treatment of IGT. In: *The Third International Conference of Diabetes; 2002; Beijing, China*; 2002.
88. Lee B, Bea J, Pyo H, et al. Biological activities of the polysaccharides produced from submerged culture of the edible Basidiomycete *Grifola frondosa*. *Enzym Microb Technol*. 2003;32(5):574–581.
89. Ohno R, Yamada K, Masaoka T, et al. A randomized trial of chemoimmunotherapy of acute nonlymphocytic leukemia in adults using a protein-bound polysaccharide preparation. *Cancer Immunol Immunother*. 1984;18(3):149–154.
90. Sakamoto J, Morita S, Oba K, Matsui T, Kobayashi M, Nakazato H. Efficacy of adjuvant immunochemotherapy with polysaccharide K for patients with curatively resected colorectal cancer: a meta-analysis of centrally randomized controlled clinical trials. *Cancer Immunol Immunother*. 2006;55(4):404–411.
91. Choi J, Kim Y, Lim H, et al. 5-fluorouracil, mitomycin-C, and polysaccharide-K adjuvant chemoimmunotherapy for locally advanced gastric cancer: the prognostic significance of frequent perineural invasion. *Hepato-Gastroenterology*. 2007;54(73): 290–297.
92. Oba K, Teramukai S, Kobayashi M, Matsui T, Kodera Y, Sakamoto J. Efficacy of adjuvant immunochemotherapy with polysaccharide K for patients with curative resections of gastric cancer. *Cancer Immunol Immunother*. 2007;56(6):905–911.
93. Huang C, Cai S, Liu J, Cai K. Clinical efficacy of PSP capsule combined chemotherapy in the treatment of advanced lung cancer. *Med Anthol*. 2000;19(6):871–872.
94. Wang B, Jia Y, Jia Y, et al. Clinical observation of synergy and attenuation of versicolor capsule in the treatment of advanced hepatocellular carcinoma. *World Chin Med*. 2009;4(4):202–203.
95. He J. Clinical manifestations of three kinds of poisonous mushroom poisoning and Ganoderma's treatment effect. *Guangdong Health Epidemic Prev Inf*. 1984;1:102–105.
96. Mao H. Clinical observation of *Ganoderma sinensis* polysaccharide in treatment of leukopenia (27 cases). *Chin J Ind Hyg Occup Dis*. 1988;14(4):251.
97. Liu L. Clinical observation of PUPS combined hepatitis B vaccine in treatment of 128 cases of chronic hepatitis B. *Clin Med*. 2009;29(6):15–16.
98. Zhang X, Wei L, Lu X. Clinical observation of PUPS in treatment of 36 cases of chronic hepatitis B. *J Xuzhou Med Coll*. 1994;14(3):240–241.
99. Liang S, Huang J. The clinical application of PUPS. *Chin J Hosp Pharm*. 1999;19(3):37–39.

100. Yan S, Cao W, Zhang Y, et al. Clinical and experimental research on *Polyporus umbellatus* polysaccharide in the treatment of chronic viral hepatitis. *Chin J Integr Med*. 1988;8(3):141–143 [131].
101. Liu Y. Recent advances in the treatment of hepatitis b with eight traditional Chinese medicines. *J Clin Med*. 2016;3(18):3693–3694.
102. Zhang L, Wang L. Effect of point injection of PPS combined with entecavir in the treatment of 23 cases of chronic hepatitis B. *Neimenggu J Tradit Chin Med*. 2016;35(8):103.
103. Wu G, Zhang L, Okuda H. Inhibitive effect of *Polyporus umbellatus* polysaccharide on cachexic manifestation induced by toxohormone-L in rats. *Chin J Integr Med*. 1997;17(4):232–233.
104. Lin Y, Wu G. Protective effect of *Polyporus umbellatus* polysaccharide on toxic hepatitis in mice. *Acta Pharmacol Sin*. 1988;9(4):345–348.
105. Yang D, Jiao Y, Du Z, Hu X, Liu F. Therapeutic effects of polyporus polysaccharide injection combined with cisplatin intraperitoneal injection on seroperitoneum due to gastric cancer. *Heb Med J*. 2018;40(5):725–731.
106. Jiang D, Zhang J, He Y, et al. Immune regulation effect of ganoderma polysaccharide capsule on immunosuppressed mice. *Med J Chin PLA*. 2016;41(9):719–724.
107. Shi M, Zhang Z, Yang Y. Antioxidant and immunoregulatory activity of *Ganoderma lucidum* polysaccharide (GLPS). *Carbohydr Polym*. 2013;95(1):200–206.
108. Yan T, Chen S, Xu L. Effect of *Ganoderma lucidum* polysaccharide on learning and memory ability and oxygen stress of model rats with Alzheimers disease. *Shaanxi Med J*. 2011;40(4):387–389. +425.
109. Pan K, Jiang Q, Liu G, Miao X, Zhong D. Optimization extraction of *Ganoderma lucidum* polysaccharides and its immunity and antioxidant activities. *Int J Biol Macromol* 2013;55:301–306.
110. Yang G, Yang L, Zhuang Y, Qian X, Shen Y. *Ganoderma lucidum* polysaccharide exerts antitumor activity via MAPK pathways in HL-60 acute leukemia cells. *J Recept Signal Transduct Res*. 2016;36(1):6–13.
111. Wang P, Zhu X, Lin Z. Antitumor and immunomodulatory effects of polysaccharides from broken-spore of *Ganoderma lucidum*. *Front Pharmacol*. 2012;3:135.
112. Gao Y, Gao H, Chan E, et al. Antitumor activity and underlying mechanisms of ganopoly, the refined polysaccharides extracted from *Ganoderma lucidum*, in mice. *Immunol Investig*. 2005;34:171–198.
113. Guo P, Sheng Y, Liu B, et al. Influence of *Ganoderma lucidum* polysaccharide on the inhibitory effects of cisplatin on the tumor growth and angiogenesis in bladder cancer (T24) cells -bearing nude mice. *Med J Chin PLA*. 2014;39(6):470–474.
114. Zhu K, Nie S, Dan S. Protective effect of polysaccharide from *Ganoderma atrum* on fasting blood glucose, serum lipids and arteriosclerotic narrowing of superior mesenteric arteries in type II diabetic rats. *Food Sci*. 2013;34(23):3004304.
115. Gong Y, Wang C, Kang J. Effects of exercise and ganoderan intervention on blood fat of rats with diabetes mellitus. *West J Tradit Chin Med*. 2013;26(8):8–10.
116. Chen Y, Chang H. Antiproliferative and differentiating effects of polysaccharide fraction from fu-ling (*Poria cocos*) on human leukemic U937 and HL-60 cells. *Food Chem Toxicol*. 2004;42(5):759–769.
117. Zhang M, Li Y, Zhang D, Zhuang P, Zhang Y. Inhibition of spontaneous lung metastasis of pachymaran on Lewis lung cancer in mice and its mechanism. *Drugs Clin*. 2013;28(6):842–846.
118. Huang Q, Jin Y, Zhang L, Cheung P, Kennedy J. Structure, molecular size and antitumor activities of polysaccharides from *Poria cocos* mycelia produced in fermenter. *Carbohydr Polym*. 2007;70(3):324–333.
119. Ke R, Lin S, Chen Y, Ji C, Shu Q. Analysis of chemical composition of polysaccharides from *Poria cocos* wolf and its anti-tumor activity by NMR spectroscopy. *Carbohydr Polym* 2010;80:31–34.

120. Lu H, Lu Y, Liu Y. Study on antioxidant properties of polysaccharides from *Poria cocos*. *Food Res Dev*. 2014;23:1–3.
121. Wei Y, Hu T, Su Z, Zeng Y, Wei X, Zhang S. Immunomodulatory and antioxidant effects of carboxymethylpachymaran on the mice infected with PCV2. *Int J Biol Macromol*. 2012;50(3):713–719.
122. Hou X, Zhang N, Xiong S, Li S, Yang B. Extraction of BaChu mushroom polysaccharides and preparation of a compound beverage. *Carbohydr Polym*. 2008;73(2): 289–294.
123. Lull C, Wichers H, Savelkoul H. Antiinflammatory and immunomodulating properties of fungal metabolites. *Mediat Inflamm*. 2005;2005(2):63–80.
124. Guo Z, Hu Y, Wang D, et al. Sulfated modification can enhance the adjuvanticity of lentinan and improve the immune effect of ND vaccine. *Vaccine*. 2009;27(5):660–665.
125. Markova N, Kussovski V, Radoucheva T, Dilova K, Georgieva N. Effects of intraperitoneal and intranasal application of lentinan on cellular response in rats. *Int Immunopharmacol*. 2002;2(12):1641–1645.
126. Drandarska I, Kussovski V, Nikolaeva S, Markova N. Combined immunomodulating effects of BCG and lentinan after intranasal application in guinea pigs. *Int Immunopharmacol*. 2005;5(4):795–803.
127. Kerekgyart C, Virag L, Tanko L, Chihara G, Fachet J. Strain differences in the cytotoxic activity and TNF production of murine macrophages stimulated by lentinan. *Int J Immunopharmacol*. 1996;18(6–7):347–353.
128. Edagawa Y, Smriga M, Nishiyama N, Saito H. Systemic administration of lentinan, a branched beta-glucan, enhances long-term potentiation in the rat dentate gyrus in vivo. *Neurosci Lett*. 2001;314(3):139–142.
129. Shi ZW, Liu Y, Xu Y, et al. Tremella polysaccharides attenuated sepsis through inhibiting abnormal $CD4^{(+)}CD25^{(high)}$ regulatory T cells in mice. *Cell Immunol*. 2014;288(1–2):60–65.
130. Li W, Liu Q, Peng H. Effects of spleen D^{Cregs} on the secretion function of $CD4^{+}$ T cells in sepsis mice after intervention of tremella polysaccharide. *Shandong Med J*. 2015;55(27):24–26.
131. Shi Z, Xu Y, Li X, Liu Q. Tremella polysaccharides attenuated sepsis through inhibiting abnormal $CD4^{+}CD25^{high}$ regulatory T cells in mice. *Chin J Immunol*. 2016;32(3):313–317.
132. Zhang X. Study on extraction and antioxidation of polysaccharides from *Tremella fuciformis*. *Liaoning Chem Ind*. 2015;9:1158–1159.
133. Zhang Z, Wang X, Zhao M, Qi H. Free-radical degradation by $Fe^{2+}/Vc/H_2O_2$ and antioxidant activity of polysaccharide from *Tremella fuciformis*. *Carbohydr Polym*. 2014;112:578–582.
134. Shen T, Duan C, Chen B, et al. *Tremella fuciformis* polysaccharide suppresses hydrogen peroxide-triggered injury of human skin fibroblasts via upregulation of SIRT1. *Mol Med Rep*. 2017;16(2):1340–1346.
135. Bach E, Costa S, Oliveira H, et al. Use of polysaccharide extracted from *Tremella fuciformis* berk for control diabetes induced in rats. *Emirates J Food Agric*. 2015;27(7):585.
136. Wang C, Meng M, Liu S, Wang L, Hou L, Cao X. A chemically sulfated polysaccharide from *Grifola frondosa* induces HepG2 cell apoptosis by notch1-NF-B pathway. *Carbohydr Polym*. 2013;95(1):282–287.
137. Zhao F, Wang Y, Song L, et al. Synergistic apoptotic effect of d-fraction from *Grifola frondosa* and vitamin C on hepatocellular carcinoma SMMC-7721 cells. *Integr Cancer Ther*. 2017;16(2):205–214.
138. Ma X, Men M, Han L, Li Z, Cao X, Wang C. Immunomodulatory activity of macromolecular polysaccharide isolated from *Grifola frondosa*. *Chin J Nat Med*. 2015;133(12):906–914.

139. Alonso E, Orozco M, Nieto A, Balogh G. Genes related to suppression of malignant phenotype induced by maitake D-fraction in breast cancer cells. *J Med Food*. 2013;16(7):602–617.
140. Klaus A, Kozarski M, Vunduk J, et al. Biological potential of extracts of the wild edible basidiomycete mushroom *Grifola frondosa*. *Food Res Int*. 2015;67:272–283.
141. Ma X, Zhou F, Chen Y, et al. A polysaccharide from *Grifola frondosa* relieves insulin resistance of HepG2 cell by Akt-GSK-3 pathway. *Glycoconj J*. 2014;31(5):355–363.
142. Xiao C, Wu Q, Xie Y, Zhang J, Tan J. Hypoglycemic effects of *Grifola frondosa* (maitake) polysaccharides F2 and F3 through improvement of insulin resistance in diabetic rats. *Food Funct*. 2015;6(11):3567–3575.
143. Ma X, Li C, Qi W, et al. Protective effect of extracellular polysaccharides from *Grifola frondosa* mycelium on CCl4-injured liver in vitro. *Bioact Carbohydr Diet Fibre*. 2015;6(1):7–14.
144. Preuss H, Echard B, Fu J, et al. Fraction SX of maitake mushroom favorably influences blood glucose levels and blood pressure in streptozotocin-induced diabetic rats. *J Med Food*. 2012;15(10):901–908.
145. Tamagawa K, Horiuchi T, Wada T, Bannai K, Ando T. Polysaccharide-K (PSK) may suppress surgical stress-induced metastasis in rat colon cancer. *Langenbeck's Arch Surg*. 2012;397(3):475–480.
146. Iino Y, Yokoe T, Maemura M, Takei H, Horguchi J, Morshita Y. A new endocrine therapy strategy for bone metastasis of breast cancer: The effect of biological response modifiers and 22-oxacarcitriol on animal models. *Breast Cancer*. 1997;4(4):311.
147. Li Q, Wang X, Chen Y, Lin J, Zhou X. Cytokines expression induced by *Ganoderma sinensis* fungal immunomodulatory proteins (FIP-gsi) in mouse spleen cells. *Appl Biochem Biotechnol*. 2010;162(5):1403–1413.
148. Ying Y, Zhang J, Tang Q, et al. Properties and in vitro immunological activities of polysaccharides from *Ganoderma sinense* fruit bodies. *Acta Edulis Fungi*. 2016;23(4): 34–39.
149. Li X, Xu W. TLR4-mediated activation of macrophages by the polysaccharide fraction from *Polyporus umbellatus*(pers.) fries. *J Ethnopharmacol*. 2011;135(1):1–6.
150. Zhang G, Zeng X, Li C, Li J, Huang Y, Han Y. Inhibition of urinary bladder carcinogenesis by aqueous extract of sclerotia of *Polyporus umbellatus* fries and polyporus polysaccharide. *Am J Chin Med*. 2011;39(1):135–144.
151. Zhao Y, Zhang L, Mao J, et al. Ergosta-4, 6, 8 (14), 22-tetraen-3-one isolated from *Polyporus umbellatus* prevents early renal injury in aristolochic acid-induced nephropathy rats. *J Pharm Pharmacol*. 2011;63(12):1581–1586.
152. Zhang G, Zeng X, Han L, Wei J, Huang H. Diuretic activity and kidney medulla AQP1, AQP2, AQP3, V2R expression of the aqueous extract of sclerotia of *Polyporus umbellatus* FRIES in normal rats. *J Ethnopharmacol*. 2010;128(2):433–437.
153. Shi Y, Liu Y, Hang L. Research on the effect of glossy ganoderma polysaccharide on anti-oxidation in exhausted mice. *Mod Prev Med*. 2006;33(6):889–890.
154. Wang Y, Wang Y, Luo Q, Zhang H, Cao J. Molecular characterization of the effects of *Ganoderma lucidum* polysaccharides on the structure and activity of bovine serum albumin. *Spectrochim Acta A Mol Biomol Spectrosc*. 2019;206(5):538–546.
155. Wang Y, Zhang M, Ruan D, et al. Chemical components and molecular mass of six polysaccharides isolated from the sclerotium of Poria cocos. *Carbohydr Res*. 2004;339(2):327–334.
156. Jin Y, Zhang L, Chen L, Cheung P, Chen L. Effect of culture media on the chemical and physical characteristics of polysaccharides isolated from *Poria cocos* mycelia. *Carbohydr Res*. 2003;338(14):1507–1515.
157. Jia X, Ma L, Li P, Chen M, He C. Prospects of *Poria cocos* polysaccharides: isolation process, structural features and bioactivities. *Trends Food Sci Technol*. 2016;54:52–62.

158. Wang X, Liu C, Dou D. Studies on the effective components of *Poria cocos* in antitumor activity. *Liaoning J Tradit Chin Med*. 2014;41:1240–1243.
159. Zhang Q, Zhang L. Studies on the antioxidative activities of ferulic acid pachymaran. *Mod Food Sci Tech*. 2011;27(9):1077–1080.
160. Cheng S, Liu Y, Mei G, Li H, Wang Y, Huang W. Studies on the antioxidative activities of carboxymethyl pachymaran. *Food R&D*. 2013;34(3):1–5.
161. Saito H, Ohki T, Takasuka N, Sasaki T. A 13C-N.M.R.-spectral study of a gel-forming, branched (1→3)-β-d-glucan, (lentinan) from lentinus edodes, and its acid-degraded fractions. Structure, and dependence of conformation on the molecular weight. *Carbohydr Res*. 1977;58(2):293–305.
162. Lu G, Fan L, Zhang Z, Pan H. Development of research on lentinan. *Acta Agriculturae Zhejiangensis*. 2009;21(2):183–188.
163. Liu J, Meng CG, Yan YH, Shan YN, Kan J, Jin CH. Structure, physical property and antioxidant activity of catechin grafted tremella fuciformis polysaccharide. *Int J Biol Macromol*. 2016;82:719–724.
164. Beats SD, Vandamme E. Extracellular tremella polysaccharides: structure, properties and applications. *Biotechnol Lett*. 2001;23:1361–1366.
165. Zhu H, Yuan Y, Liu J, Zheng L, Chen L, Ma A. Comparing the sugar profiles and primary structures of alkali-extracted water-soluble polysaccharides in cell wall between the yeast and mycelial phases from *Tremella fuciformis*. *J Microbiol*. 2016;54(5):381–386.
166. Huang J, Jia F, Shi W. Research development of *Tremella fuciformis* polysaccharides. *Chongqing J Res Chin Drugs Herbs*. 2013;1:43–47.
167. Zhang L, Li H, Bai H. Comparison of the determination of polysaccharide with 3,5-dinitrosalicylic acid and phenol sulfate in tremella polysaccharide enteric-coated capsules. *Tianjin Pharm*. 2008;20(3):14–17.
168. Wei H, Zhang D, Lu L, Chen F. Study on extraction and antioxidation of polysaccharides from several edible fungi. *J Nanjing Norm Uni*. 2017;40(2):72–76.
169. Guo X, He G, Liu W, Yang Y, Liu B, Gou X. Preparation and antioxidative activities of the sulfated tremella polysaccharides. *Nat Prod Res Dev*. 2013;25:747–751.
170. Wen L, Gao Q, Ma C, et al. Effect of polysaccharides from *Tremella fuciformis* on UV-induced photoaging. *J Funct Foods*. 2016;20:400–410.
171. Zhang Z, Sun D, Xu M, Tian H, Pan J. Study on the antioxidation effect of tremella polysaccharide. *Food Res Dev*. 2014;35(18):10–15.
172. Kim J, Ha H, Lee M, et al. Effect of *Tremella fuciformis* on the neurite outgrowth of PC12h cells and the improvement of memory in rats. *Biol Pharm Bull*. 2007;30(4):708–714.
173. Park K, Lee S, Kim H, Yamazaki M, Chiba K, Ha H. The neuroprotective and neurotrophic effects of *Tremella fuciformis* in PC12h cells. *Mycobiology*. 2007;35(1):11–15.
174. Hu X, Liu Y, Wu J. Protective effect of polysaccharides from *Tremella fuciformis* on apoptosis of differentiated PC12 cells induced by MPP^+. In: *Paper Presented at the 18th National Conference on Neurology of the Chinese Medical Association 2015*; 2015.
175. He X, Wang X, Fang J, et al. Polysaccharides in *Grifola frondosa* mushroom and their health promoting properties: a review. *Int J Biol Macromol*. 2017;101:910–921.
176. Rau U, Kuenz A, Wray V, Nimtz M, Wrenger J, Cicek H. Production and structural analysis of the polysaccharide secreted by trametes (Coriolus) versicolor ATCC 200801. *Appl Microbiol Biotechnol*. 2009;81(5):827–837.
177. Santos Arteiro JM, Martins MR, Salvador C, Candeias MF, Karmali A, Caldeira AT. Protein-polysaccharides of trametes versicolor: production and biological activities. *Med Chem Res*. 2012;21(6):937–943.
178. Chang Y, Zhang M, Jiang Y, et al. Preclinical and clinical studies of *Coriolus versicolor* polysaccharopeptide as an immunotherapeutic in China. *Discov Med*. 2017;23(127):207–219.

179. Zhang W, Xing L, Wang Y, Wang Y. Clinical observation of Dong Fang Yunzhi capsules in the treatment of viral hepatitis. *Shanghai J Tradit Chin Med*. 1994;9:35–36.
180. Rao G, Tang X. Clinical observation of versicolor intracellular polysaccharide in the treatment hyperlipidemia. *Chongqing Med J*. 2007;36(13):1306–1307.
181. Hu Y, Jin R, Fang J, Mao Q, Qi L, Dai S. The anti-ulcer effect of *Coriolus versicolor* polysaccharopeptide. *Chin Tradit Pat Med*. 1990;12(11):22–23.
182. Maehara Y, Tsujitani S, Saeki H, et al. Biological mechanism and clinical effect of protein-bound polysaccharide K (KRESTIN(R)): review of development and future perspectives. *Surg Today*. 2012;42(1):8–28.
183. Nie S, Zhang H, Li W, Xie M. Current development of polysaccharides from ganoderma: isolation, structure and bioactivities. *Bioact Carbohydr Diet Fibre*. 2013;1(1):10–20.
184. Xu S, Kuang Y, Jiang W, Ma J, Wu X, Jin P. Advances on chemical constituents and bioactivities of polyporus. *Cent South Pharm*. 2016;14(7):746–751.